

hw0526

16.1(a)

Given the logit model:

$$P(AUTO = 1) = \Lambda(-0.2376 + 0.5311DTIME)$$

When $DTIME=1$:

$$z = -0.2376 + 0.5311(1) = 0.2935$$

Therefore,

$$\hat{P}(AUTO = 1 \mid DTIME = 1) = \frac{1}{1+e^{-0.2935}} = 0.5729$$

The estimated probability that a person chooses automobile transportation is approximately **0.573**, or **57.3%**.

16.1(b)

Using the probit model from Example 16.4:

$$P(AUTO = 1) = \Phi(-0.0644 + 0.3000DTIME)$$

When $DTIME=1$:

$$z = -0.0644 + 0.3000(1) = 0.2356$$

Therefore,

$$\hat{P}(AUTO = 1 \mid DTIME = 1) = \Phi(0.2356) \approx 0.5932$$

The estimated probability that a person chooses automobile transportation is approximately **0.593**, or **59.3%**. This is slightly higher than the logit estimate of **0.573**, but the two estimates are very close.

16.1(c)

For the logit model, the marginal effect is:

$$\frac{\partial P}{\partial DTIME} = \Lambda(z)[1 - \Lambda(z)]\gamma_2$$

When bus travel time is 30 minutes longer than auto travel time, $DTIME=3$.

$$z = -0.2376 + 0.5311(3) = 1.3557$$

$$\Lambda(1.3557) \approx 0.7951$$

Thus,

$$\frac{\partial P}{\partial DTIME} = 0.7951(1 - 0.7951)(0.5311) \approx 0.0865$$

Using the linear probability model from Example 16.2:

$$\widehat{AUTO} = 0.4848 + 0.0703DTIME$$

The marginal effect is simply:

$$0.0703$$

The logit marginal effect is approximately **0.0865**, while the linear probability model marginal effect is **0.0703**. The logit estimate is larger, but both imply that a 10-minute increase in bus travel time relative to auto travel time increases the probability of choosing automobile transportation.

16.1(d)

Since $DTIME = (BUSTIME - AUTOTIME)/10$, if driving takes 50 minutes longer than bus travel, then:

$$DTIME = -5$$

For the logit model:

$$z = -0.2376 + 0.5311(-5) = -2.8931$$

$$\Lambda(-2.8931) \approx 0.0525$$

The logit marginal effect is:

$$\begin{aligned} \frac{\partial P}{\partial DTIME} &= \Lambda(z)[1 - \Lambda(z)]\gamma_2 \\ &= 0.0525(1 - 0.0525)(0.5311) \approx 0.0264 \end{aligned}$$

For the probit model from Example 16.4:

$$P(AUTO = 1) = \Phi(-0.0644 + 0.3000DTIME)$$

$$z = -0.0644 + 0.3000(-5) = -1.5644$$

The probit marginal effect is:

$$\frac{\partial P}{\partial DTIME} = \phi(z)\beta_2$$

$$= \phi(-1.5644)(0.3000) \approx 0.1175(0.3000) = 0.0353$$

The logit marginal effect is approximately **0.0264**, while the probit marginal effect is approximately **0.0353**. The probit estimate is slightly larger, but both imply that a 10-minute decrease in driving time increases the probability of choosing automobile transportation.

16.4(a)

Given the logit model:

$$P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$$

Using the maximum likelihood estimates:

$$\hat{\gamma}_1 + \hat{\gamma}_2 x = -1.836 + 3.021x$$

When $x=1.5$:

$$z = -1.836 + 3.021(1.5) = 2.6955$$

Therefore,

$$\hat{P}(y = 1 \mid x = 1.5) = \frac{1}{1+e^{-2.6955}} \approx 0.9368$$

The estimated probability that $y = 1$ when $x = 1.5$ is approximately **0.937**, or **93.7%**.

16.4(b)

From part (a):

$$\hat{P}(y = 1 \mid x = 1.5) \approx 0.9368$$

Since:

$$0.9368 > 0.5$$

we predict:

$$\hat{y} = 1$$

The predicted value is $\hat{y} = 1$. This matches the actual outcome of the first observation, which is $y = 1$, so the prediction is correct.

16.4(c)

The three observations from Example 16.3 are:

$$(y_1, y_2, y_3) = (1, 1, 0)$$

$$(x_1, x_2, x_3) = (1.5, 0.6, 0.7)$$

For the logit model:

$$P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$$

Using $\gamma_1 = -1$ and $\gamma_2 = 2$:

$$z_1 = -1 + 2(1.5) = 2, \text{ so } \Lambda(2) = 0.8808$$

$$z_2 = -1 + 2(0.6) = 0.2, \text{ so } \Lambda(0.2) = 0.5498$$

$$z_3 = -1 + 2(0.7) = 0.4, \text{ so } 1 - \Lambda(0.4) = 0.4013$$

Thus,

$$L(-1, 2) = 0.8808(0.5498)(0.4013) \approx 0.1944$$

The maximized log-likelihood is given as:

$$\log L(\hat{\gamma}_1, \hat{\gamma}_2) = -1.612$$

Therefore,

$$L(\hat{\gamma}_1, \hat{\gamma}_2) = e^{-1.612} \approx 0.1995$$

The likelihood value at $(\gamma_1, \gamma_2) = (-1, 2)$ is approximately **0.1944**. The likelihood value at the maximum likelihood estimates is approximately **0.1995**, which is larger. This is expected because the MLE maximizes the likelihood function.

16.4(d)

Statement: For the logit model, the value of the likelihood function is always between zero and one.

Answer: True.

For each observation, the logit model gives a probability:

$$0 < \Lambda(z) < 1$$

and:

$$0 < 1 - \Lambda(z) < 1$$

The likelihood function is the product of these probabilities:

$$L(\gamma_1, \gamma_2) = \prod_{i=1}^N P(y_i = 1)^{y_i} [1 - P(y_i = 1)]^{1-y_i}$$

Since each term is a probability between 0 and 1, their product is also between 0 and 1.

Therefore:

$$0 < L(\gamma_1, \gamma_2) < 1$$

16.4(e)

Statement: For the logit model, the value of the log-likelihood function is always negative.

Answer: True.

From part (d), the likelihood function is between 0 and 1:

$$0 < L(\gamma_1, \gamma_2) < 1$$

The log-likelihood function is:

$$\log L(\gamma_1, \gamma_2)$$

Since the natural logarithm of any number between 0 and 1 is negative:

$$\log L(\gamma_1, \gamma_2) < 0$$

Therefore, the log-likelihood value for the logit model is negative.

16.18(a)

The linear probability model is estimated as:

$$DELINQUENT = \beta_0 + \beta_1 LVR + \beta_2 REF + \beta_3 INSUR + \beta_4 RATE + \beta_5 AMOUNT + \beta_6 CREDIT + \beta_7 TERM + \beta_8 ARM + e$$

Using White heteroskedasticity-robust standard errors, the significant variables at the 5% level are:

LVR, *REF*, *INSUR*, *RATE*, *CREDIT*, *TERM*, and *ARM*.

The variable *AMOUNT* is not statistically significant at the 5% level.

The signs of *LVR*, *RATE*, *CREDIT*, and *ARM* are reasonable. A higher loan-to-value ratio and a higher interest rate increase the probability of delinquency. A higher credit score lowers the probability of delinquency. Adjustable-rate mortgages also increase the probability of delinquency.

The negative signs on *REF*, *INSUR*, and *TERM* are less obvious, but they may reflect differences in borrower characteristics or lower payment burdens.

Overall, most coefficient signs are reasonable. The LPM suggests that delinquency is more likely when *LVR*, *RATE*, and *ARM* are higher, and less likely when *CREDIT* is higher.

16.18(b)

The logit model is estimated as:

$$P(\text{DELINQUENT} = 1) = \Lambda(\beta_0 + \beta_1 LVR + \beta_2 REF + \beta_3 INSUR + \beta_4 RATE + \beta_5 AMOUNT + \beta_6 CREDIT + \beta_7 TERM + \beta_8 ARM)$$

The signs of all estimated coefficients are the same as in the linear probability model:

$$LVR > 0, REF < 0, INSUR < 0, RATE > 0, AMOUNT > 0, CREDIT < 0, TERM < 0, \text{ and } ARM > 0.$$

At the 5% significance level, the significant variables in the logit model are:

REF, *INSUR*, *RATE*, *TERM*, and *ARM*.

LVR, *AMOUNT*, and *CREDIT* are not significant at the 5% level, although they are close to being significant at the 10% level.

The signs of the logit estimates are the same as those in the linear probability model. However, the significance is not exactly the same. In the *LPM*, *LVR* and *CREDIT* were significant at the 5% level, but in the logit model they are not significant at the 5% level.

16.18(c)

The predicted probabilities for observations 500 and 1000 are:

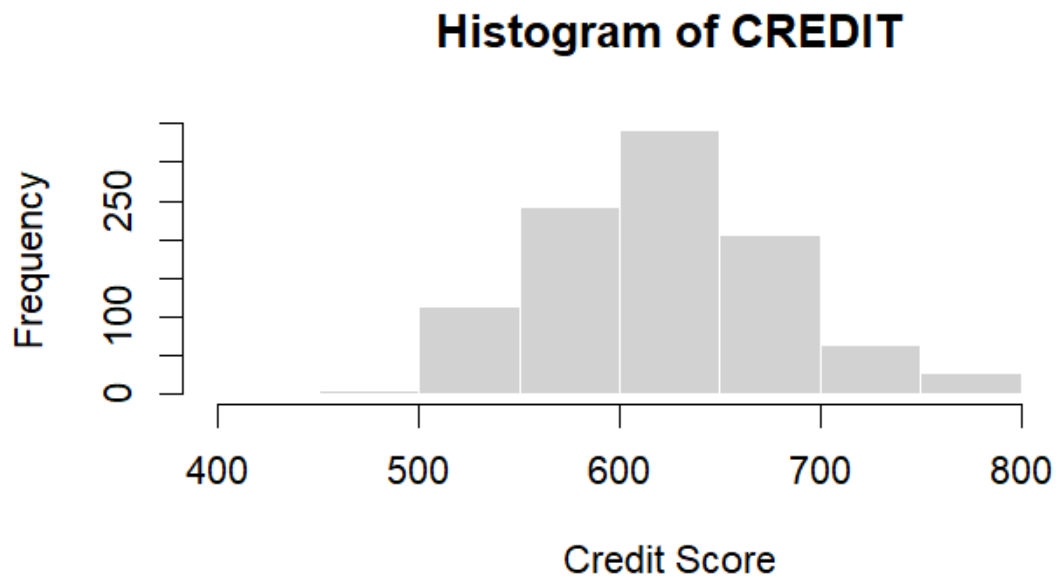
Observation	Actual DELINQUENT	LPM Predicted Probability	Logit Predicted Probability
500	0	0.1828	0.1336
1000	0	0.5785	0.6266

For observation 500, both predicted probabilities are below 0.5, so both models predict DELINQUENT=0. This matches the actual outcome.

For observation 1000, both predicted probabilities are above 0.5, so both models predict DELINQUENT=1. This does not match the actual outcome, since the actual value is DELINQUENT=0.

Both models correctly classify observation 500 but incorrectly classify observation 1000. The logit model gives a lower predicted probability than the LPM for observation 500, but a higher predicted probability than the LPM for observation 1000.

16.18(d)



The predicted probabilities of delinquency for different credit scores are:

CREDIT	LPM Predicted Probability	Logit Predicted Probability
500	0.5533	0.4967
600	0.5091	0.4086
700	0.4649	0.3261

Both models show that a higher credit score lowers the predicted probability of delinquency.

In the linear probability model, the effect of *CREDIT* is constant. When *CREDIT* increases by 100 points, the predicted probability falls by about 0.0442.

In the logit model, the effect is nonlinear. The predicted probability falls from 0.4967 to 0.4086 when *CREDIT* rises from 500 to 600, and from 0.4086 to 0.3261 when *CREDIT* rises from 600 to 700.

The two models give the same qualitative result: higher credit scores reduce delinquency risk. However, the LPM gives higher predicted probabilities and a

constant change, while the logit model gives lower predicted probabilities and a nonlinear change.

16.18(e)

The marginal effects of *CREDIT* are:

CREDIT	LPM Marginal Effect	Logit Predicted Probability	Logit Marginal Effect
500	-0.0004419	0.4967	-0.0008911
600	-0.0004419	0.4086	-0.0008613
700	-0.0004419	0.3261	-0.0007833

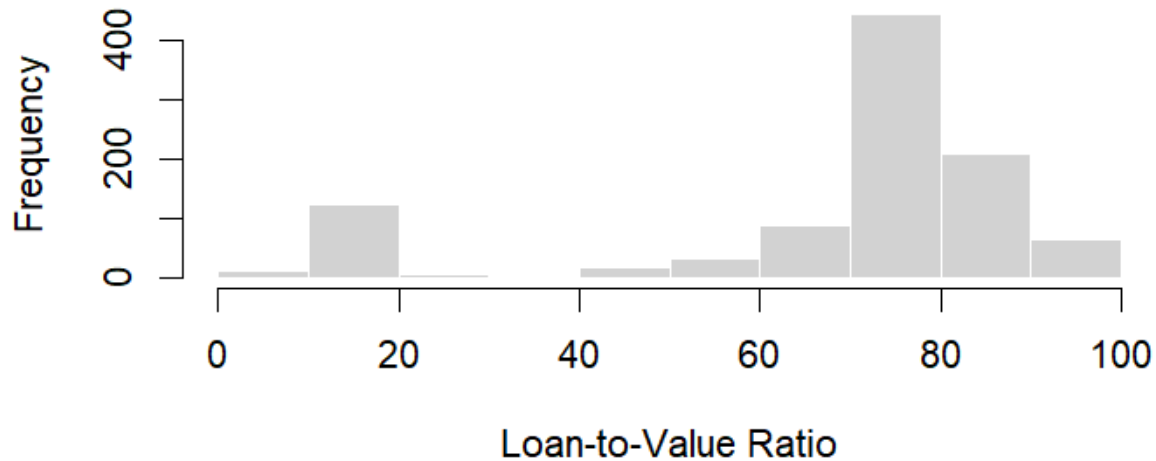
In the linear probability model, the marginal effect of *CREDIT* is constant. A one-point increase in credit score lowers the probability of delinquency by about 0.000442, or about 0.044 percentage points. A 100-point increase lowers the probability by about 4.42 percentage points.

In the logit model, the marginal effect depends on the predicted probability. A one-point increase in credit score lowers the probability of delinquency by about 0.000891 when *CREDIT*=500, 0.000861 when *CREDIT*=600, and 0.000783 when *CREDIT*=700.

Both models imply that higher credit scores reduce the probability of delinquency. However, the LPM marginal effect is constant, while the logit marginal effect varies with the level of credit score and is larger in absolute value when the credit score is lower.

16.18(f)

Histogram of LVR



The predicted probabilities of delinquency for different loan-to-value ratios are:

LVR	LPM Predicted Probability	Logit Predicted Probability
20	0.4117	0.2349
80	0.5091	0.4086

Both models show that a higher loan-to-value ratio increases the predicted probability of delinquency.

In the linear probability model, when LVR increases from 20 to 80, the predicted probability rises from 0.4117 to 0.5091. This is an increase of about 0.0974, or 9.74 percentage points.

In the logit model, when LVR increases from 20 to 80, the predicted probability rises from 0.2349 to 0.4086. This is an increase of about 0.1737, or 17.37 percentage points.

Both models imply that a higher LVR increases delinquency risk. However, the logit model predicts a larger increase in delinquency probability than the LPM.

16.18(g)

Using a cutoff value of 0.5, the percent correctly predicted is:

Model	Percent Correct
Linear Probability Model	85.8%
Logit Model	85.4%

For the linear probability model, the confusion matrix shows that 727 non-delinquent loans and 131 delinquent loans are correctly predicted. Therefore, the total number of correct predictions is:

$$727 + 131 = 858$$

So the percent correctly predicted is:

$$858/1000 = 85.8\%$$

For the logit model, the confusion matrix shows that 736 non-delinquent loans and 118 delinquent loans are correctly predicted. Therefore, the total number of correct predictions is:

$$736 + 118 = 854$$

So the percent correctly predicted is:

$$854/1000 = 85.4\%$$

The linear probability model has a slightly higher overall prediction accuracy than the logit model, 85.8% compared with 85.4%. However, the difference is very small. The LPM predicts more delinquent loans correctly, while the logit model predicts more non-delinquent loans correctly.

16.18(h)

The logit model is estimated using the first 500 observations and then used to predict delinquency for observations 501 to 1000.

Using a cutoff of 0.5, the confusion matrix is:

Actual \ Predicted	0	1
Actual 0	406	27
Actual 1	33	34

The percent correctly predicted is:

$$(406 + 34)/500 = 88\%$$

Although the 0.5 cutoff gives a high overall prediction accuracy, it is not necessarily the best rule for a lender. The lender's objective is to make loans to borrowers who will repay on schedule, so the relevant issue is the delinquency rate among approved borrowers.

Using different approval cutoffs gives the following results:

Cutoff	Number Approved	Approval Rate	Delinquency Rate Among Approved
0.05	108	21.6%	3.70%
0.10	262	52.4%	4.20%
0.20	379	75.8%	5.54%
0.30	406	81.2%	5.67%
0.50	439	87.8%	7.52%

A lower cutoff reduces the delinquency rate among approved borrowers, but it also reduces the number of loans approved. For example, a cutoff of 0.10 approves 262 loans with a delinquency rate of only 4.20%, while a cutoff of 0.50 approves 439 loans with a delinquency rate of 7.52%.

The 0.5 cutoff is not necessarily the best decision rule for a lender. The cutoff should depend on the cost of delinquency and the benefit from making a loan. A more conservative lender should use a lower cutoff, such as 0.10 or 0.20, to reduce the delinquency rate among approved borrowers.